



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 2

SECP2523

SECTION 02

DATABASE

LECTURER: DR. SEAH CHOON SEN

NO	NAME	MATRIC NO
1	YASMIN BATRISYIA BINTI ZAHIRUDIDN	A23CS0201

Assignment 2 SECPH 2523

QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offences by students in a college.

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
				00-690	15/03/06	1	20.00
				00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
				00-770	19/06/06	2	30.00

- a) Table 2.1 is said to be unnormalized table. Why?

It contains redundant data, such as repeated information about the same student (e.g., StudentID, Name, LicenseNo, and CarReg) for different offences.

- b) Transform the unnormalized Table 2.1 into a First Normal Form (1NF) table. Show the transformed 1 NF table. What is the primary key for this new 1 NF table.

1 NF

StudentFine

<u>StudentID</u>	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
F1011	CK Fong	P100	JGK9201	00-690	15/03/06	1	20.00
F1011	CK Fong	P100	JGK9201	00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
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QUESTION 2

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
		12	Pauline James	B	\$50
		16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
		17	Monique Williams	B	\$50

1NF

<u>Project number</u>	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
1023	Madagascar travel site	12	Pauline James	B	\$50
1023	Madagascar travel site	16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
1056	Online estate agency	17	Monique Williams	B	\$50

2 NF

Project

<u>Project number</u>	Project name
1023	Madagascar travel site
1056	Online estate agency

Employee

<u>Employee number</u>	Employee name	Rate category	Hourly rate
11	Vincent Radebe	A	\$60
12	Pauline James	B	\$50
16	Charles Ramoraz	C	\$40
17	Monique Williams	B	\$50

3 NF

Project

<u>Project number</u>	Project name
1023	Madagascar travel site
1056	Online estate agency

Employee

<u>Employee number</u>	Employee name
11	Vincent Radebe
12	Pauline James
16	Charles Ramoraz
17	Monique Williams

Rate

<u>Rate category</u>	Hourly rate
A	\$60
B	\$50
C	\$40

EmployeeDetails

<u>Project number</u>	<u>Employee number</u>	Rate category
1023	11	A
1023	12	B
1023	16	C
1056	11	A
1056	17	B